# Nattanitcha Promboon

PRODUCT DESIGN & PROTOTYPING · CO-OP INTERN

nattanitchafarm@gmail.com
Portfolio

**Full-time co-op · 13 Jan – 28 May 2027**

## EDUCATION

**King Mongkut's University of Technology Thonburi (KMUTT)**
B.Sc. Media Technology
2023 – present · co-op track
**GPA 3.61 / 4.00**

**Wittayanukulnaree School**
Science, Mathematics, Technology & Environment program

## SKILLS

**Design / Prototyping**
Sketching & illustration · UI/motion · sewing & embroidery · assembly & soldering · CAD & 3D printing

**Electronics**
Wiring & soldering · ESP32, XIAO ESP32S3 Sense, Arduino · motor drivers & servos · Hall, ultrasonic & light sensors · OLED & LED strips · 18650 + BMS, USB-C

**Tools**
Figma · Tinkercad · SolidWorks (learning) · Blender · Maya · Photoshop · Illustrator · Creality Print · VS Code · Office

**AI-assisted coding**
I direct AI coding agents (Claude Code, Codex, Antigravity) with written specs. A second AI reviews code and test results. I verify results.

## AWARDS & ACTIVITIES

**1st Place**: KMUTT Deep Sea Jam Hackathon 2026 (DeepCore)

**Staff**: FANTASTECH Media Exhibition 2026, KMUTT

**Freelance (ongoing)**: commission illustration & handmade dolls

## LANGUAGES

Thai: native
English: working proficiency (technical reading & writing)

## PROFILE

Design and build physical prototypes as a fourth-year Media Technology student, using CAD, my own 3D printer, wiring and soldering. Have led every team project and revise designs after failed tests. Set specifications, direct AI to write most of the code and check results.

## SELECTED PROJECTS

**METO:** small interactive robot for learning block coding — Research → thesis · 2025–present

- Lead a 3-person thesis team comparing beginner learning with and without robot expressions. A study with first-year KMUTT students is planned, with ethics documents in preparation. The camera reads facial expressions so the robot can turn towards learners and mirror their expressions on screen.
- Designed the screens and drew the 62-piece robot avatar for the local "METO Robot" block-coding web prototype. Teacher and student roles are set by class. Progress counts only after code runs without errors. The AI helper gives hints.
- Designed a five-layer trapezoid body so the arms can close in and grip, with easy assembly without heat-set inserts. The Apr 2026 version's printed oval arms with anti-slip bumps lifted a cup on USB power. The current tracked version (about 25–30 cm tall, 20–25 cm wide, 30–35 cm deep) drives wirelessly and displays its face on two 18650 cells with battery protection and USB-C charging. Its arm servos are not connected yet. The first prototype used car wheels. The printed tracks turn poorly, so I plan two wheels and a steel ball caster.

**Magnetic Universe:** magnetic LED wall panel — Jan – May 2026 · course project · team of 4

- Designed and built a 400 × 400 mm light panel for two magnet pens, with pools of light that blend together. Specified position tracking from the strongest of 16 Hall sensors and its nearby sensors, weighted by signal strength, rather than averaging the whole panel.
- Designed and printed threaded rods with twist-on caps, finding that the threads were too thin to print. Added heat-resistant fibre fill to diffuse visible LED dots without thickening the 6 cm frame, which weighs 3–4 kg and stands at 60°.

**DeepCore → DeepSphere:** deep-sea device — 2026 – present · 1st Place · team lead

- Chose a physical device over a game, map or dashboard. DeepCore, the hackathon winner, uses an ESP32 controller assembled from real parts to drive a live 3D web scene featuring 44 real species backed by 10 peer-reviewed sources.
- Defined part dimensions, thin walls, support-free printing and snap-fit joints for DeepSphere, avoiding heat-set inserts. Modelled v1 myself and made v2 CAD with AI help. The six-screen ring is built, but most screens need rewiring. The three-layer sphere remains a long-term concept.

**MOFYCH Commission Website:** service design — honeykung.github.io/commission-tag

- Designed the whole site in Figma as "art + web", with my hand-drawn clouds, logo and animation. Added email opening alerts, email-confirmed cancellations and a Google Apps Script queue sheet. The matcher covers 1,131 paints from 5 brands. Transcribed most charts and checked the rest against published data. Set a colour-difference target of $\Delta E < 5$ (lower is closer) and excluded fluorescent and metallic paints from two-paint recipes. AI implemented the standard CIEDE2000 formulas from my requirements.

**Digital Puzzle Box** — prototype

- Redesigned a flippable puzzle box as a flat board when its gyro failed. Players find a real hidden key to complete the on-screen puzzle.

**Smaller builds** — self-initiated

- Designed and 3D-printed a 10 × 10 × 15.4 cm **Minecraft Pixel Lamp** in Tinkercad. Specified an ESP32 flame effect that steps yellow → orange → red from bottom to top, with code limiting brightness and current (5 V, 1.2 A) for phone-charger power.
- Designed **svgtostl** to turn drawings into printable keychains and sewing templates for tracing and cutting fabric. Required sharp corners and accurate curves, rejecting point reduction that visibly faceted curves at 0.2 mm nozzle scale. AI implemented the code.
- Sewed **cotton dolls** of my original characters from purchased/public patterns, using my character sheets, embroidered faces and removable magnetic ears. Purchased patterns bar licensed characters. Am developing my own pattern in Blender.
- Designed the concept for **OC Companion**, a device about 4–5 cm per side with an AMOLED screen face and figure-style sculpted hair. Plan desk, portable and keychain use, with each device made to order for a buyer's original character. Hardware is not yet built.
- Designed and printed an 18-well **mini paint palette** in a 109 × 75 mm PLA tray, with a side channel for test mixing. It is in use.
- Prepared four **MOFYCH collector card** designs for factory printing, four cards each at 90 × 50 mm. Separated artwork into holographic foil, matte and smooth-finish layers.